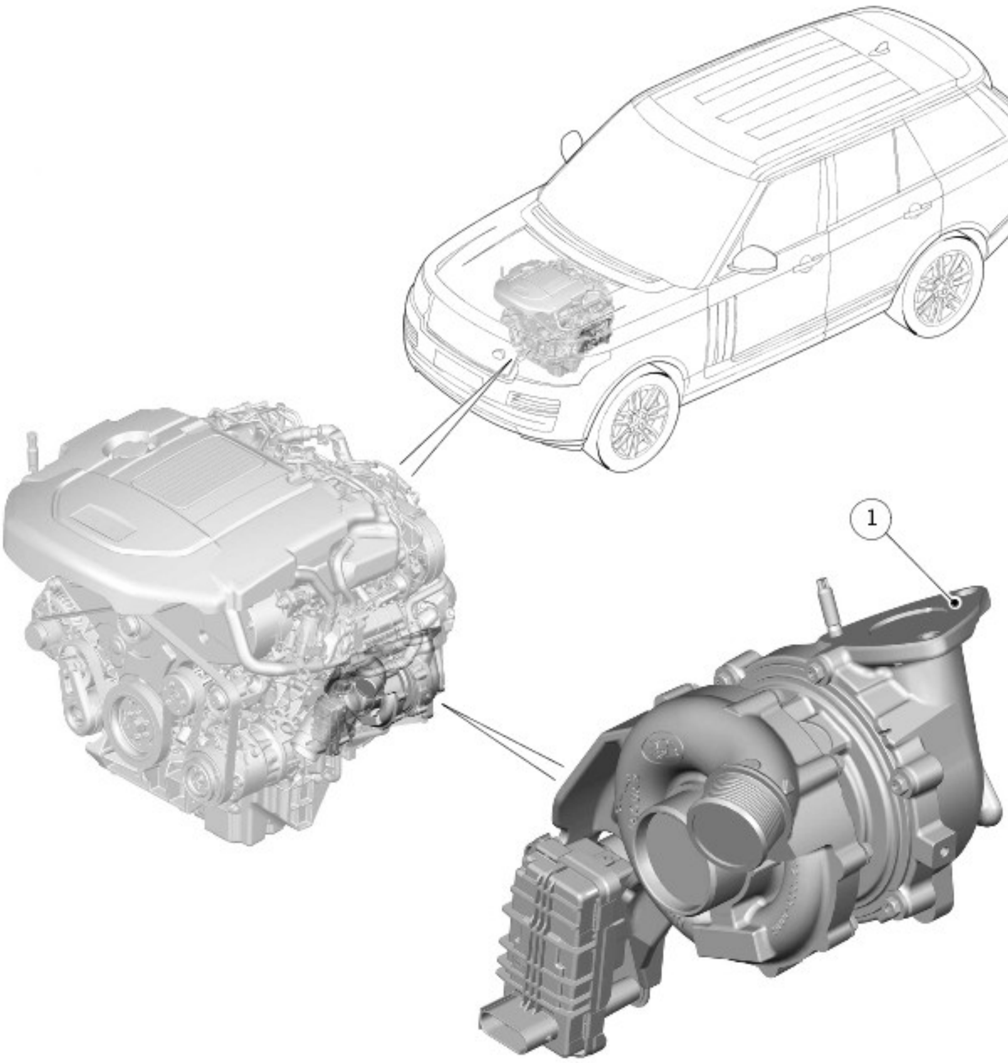
2016.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS – TURBOCHARGER – TDV6 3.0L DIESEL

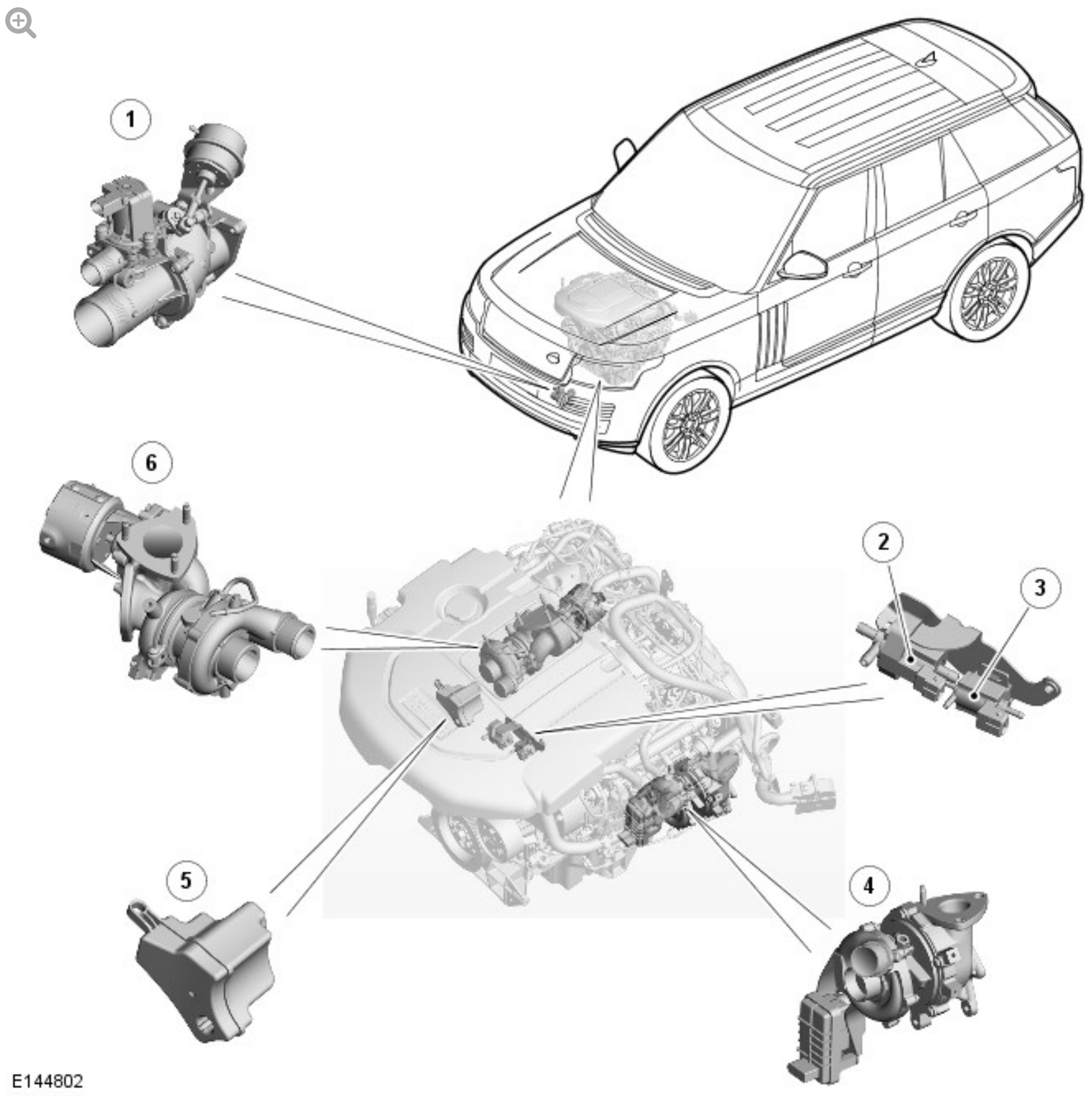
DESCRIPTION AND OPERATION

COMPONENT LOCATION – SHEET 1 OF 2 – VEHICLES WITH SINGLE TURBOCHARGER



ITEM	DESCRIPTION
1	Variable Geometry Turbine (VGT) turbocharger

COMPONENT LOCATION - SHEET 2 OF 2 - VEHICLES WITH TWIN TURBOCHARGER



ITEM	DESCRIPTION
1	Charge air intake valve
2	Turbine intake shut-off valve solenoid
3	Charge air solenoid
4	Variable Geometry Turbine (VGT) turbocharger - Primary
5	Vacuum reservoir
6	Fixed vane turbocharger - Secondary

OVERVIEW

Depending on the market, the TDV6 3.0L diesel engine is available in two variants. The single turbocharger variant uses a single Variable Geometry Turbine (VGT) type turbocharger, the twin turbocharger variant uses two turbochargers. The twin turbocharger variant comprises a VGT type turbocharger (primary turbocharger), and a fixed vane type (secondary turbocharger) turbocharger.

On the twin turbocharger variant, both turbochargers are used in a parallel sequential turbocharging system, which enables the engine to achieve quick throttle response at low engine speeds and efficient use of exhaust gas energy at high engine speeds. The primary turbocharger operates through the entire engine speed range, but is at its most efficient, at engine speeds of up to 2800 RPM it is running in a mono turbocharger mode. At engine speeds above 2800 RPM under load, the secondary turbocharger comes into operation, with both of the turbochargers running in a parallel bi-turbocharger mode.

The twin turbocharger system comprises the two turbochargers, the turbine intake shut-off valve solenoid, and the charge air intake valve, which incorporates the charge air recirculation solenoid and the charge air shut-off valve. The operation of the system is controlled by the Engine Control Module (ECM).

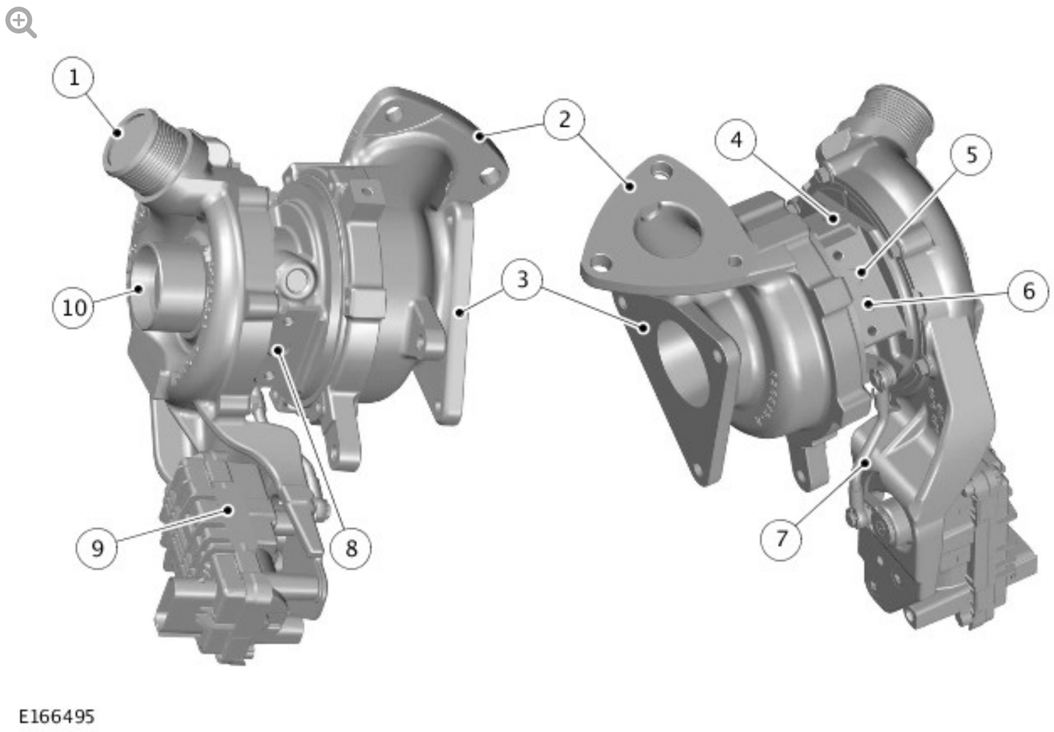
The turbochargers and the charge air intake valve are interconnected by the charge air ducts of the intake air distribution and filtering system.
For additional information, refer to: Intake Air Distribution and Filtering (303-12A Intake Air Distribution and Filtering - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

DESCRIPTION

Each turbocharger consists of two turbo elements, a turbine wheel and compressor wheel, enclosed separately in cast housings and mounted on a common shaft. The shaft in the primary turbocharger rotates in two ball bearings, however the secondary turbocharger comprises two semi-floating bearings which provide the shaft rotation.

The turbocharger bearings are lubricated and cooled by an oil supply from the cylinder block. The primary turbocharger has a connection for the engine cooling circuit to provide the appropriate cooling efficiency of the ball bearing. The primary turbocharger return oil drains into a dedicated area of the oil pan, assisted by an oil scavenge element in the vacuum pump. If fitted, the secondary turbocharger return oil drains into a dedicated area of the crankcase directly via drain tube.
For additional information, refer to: Engine (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation) / Engine (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).

VARIABLE GEOMETRY TURBINE (VGT) TURBOCHARGER - PRIMARY



ITEM	DESCRIPTION
1	Charge air outlet
2	Exhaust inlet
3	Exhaust outlet
4	Oil feed connection
5	Coolant return connection

6	Coolant supply connection
7	Variable Geometry Turbine (VGT) vane actuator rod
8	Oil return connection
9	Variable Geometry Turbine (VGT) vane actuator
10	Clean air inlet

The primary turbocharger is attached to the left exhaust manifold and secured to 3 studs on a flange on the manifold with nuts. On production, no gasket is used to seal the joint surface between the turbocharger and the manifold. In-service vehicles will require a service gasket to be fitted if the joint surface is disturbed.

A second flange on the turbocharger provides for the attachment of the left down pipe of the exhaust system. Three screws secure the down pipe to the flange and a gasket seals the joint.

The compressor end of the turbocharger has two hose connections. The central hose connection provides a clean air supply from the air cleaner to the compressor. The second connection on the outside of the housing provides for the attachment of the charge air ducts.

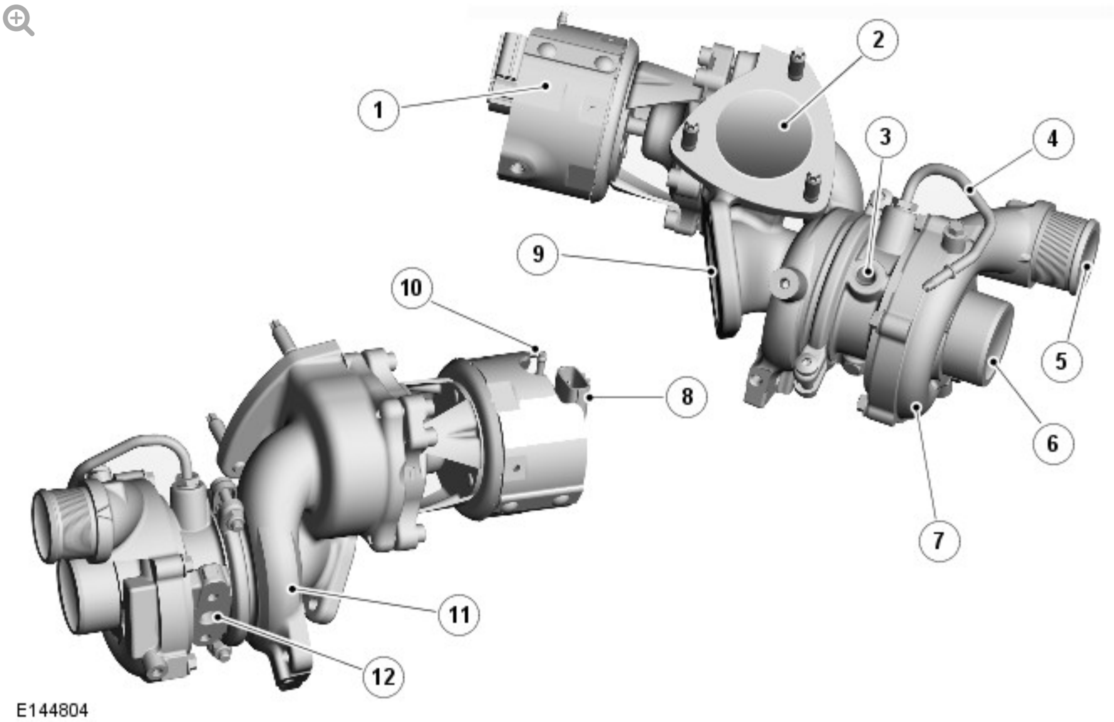
The turbocharger comprises a turbine wheel and a machined compressor wheel, sharing a common shaft which is supported on ball bearings. The application of the machined compressor wheel and the ball bearing results improved durability reduced Noise, Vibration and Harshness (NVH) and optimized fuel economy. The turbocharger receives an engine oil feed via a pipe from the cylinder block. The pipe supplies the turbocharger with engine oil for lubrication purposes. An oil drain pipe from the turbocharger allows oil to drain from the turbocharger into the cylinder block. The turbocharger has a connection for the engine cooling circuit to provide the appropriate cooling efficiency of the ball bearing.

An integral bracket houses the VGT vane actuator. The VGT vane actuator is connected to an eccentric lever which moves the turbine housing to adjust the position of the vanes. When the actuator operates a boss is rotated, which in turn moves the lever and changes rotary motion into linear motion. The lever is connected to a rod attached to the outside of the turbine housing and the linear motion is converted back to rotary motion of the housing.

Operation of the VGT vane actuator is controlled by the ECM. The VGT vane actuator also provides the ECM with a feedback signal to determine the pitch angle of the vanes.

The variable vanes in the turbocharger improve the exhaust gas power transfer to the turbine wheel which in turn drives the compressor wheel. This greatly assists the increase in charge air pressure at low engine speeds.

SECONDARY TURBOCHARGER (IF EQUIPPED)



ITEM	DESCRIPTION
1	Turbine intake shut-off valve
2	Exhaust intake
3	Oil feed connection
4	Seal pressurizing air pipe

5	Charge air outlet
6	Clean air intake
7	Compressor housing
8	Electrical connector - Turbine intake shut-off valve position sensor
9	Exhaust outlet
10	Vacuum connection
11	Turbine housing
12	Oil return connection

The secondary turbocharger is attached to the right exhaust manifold and is secured to 3 studs on a flange on the manifold with nuts. On production, no gasket is used to seal the joint surface between the turbocharger and the manifold. In-service vehicles will require a service gasket to be fitted if the joint surface is disturbed.

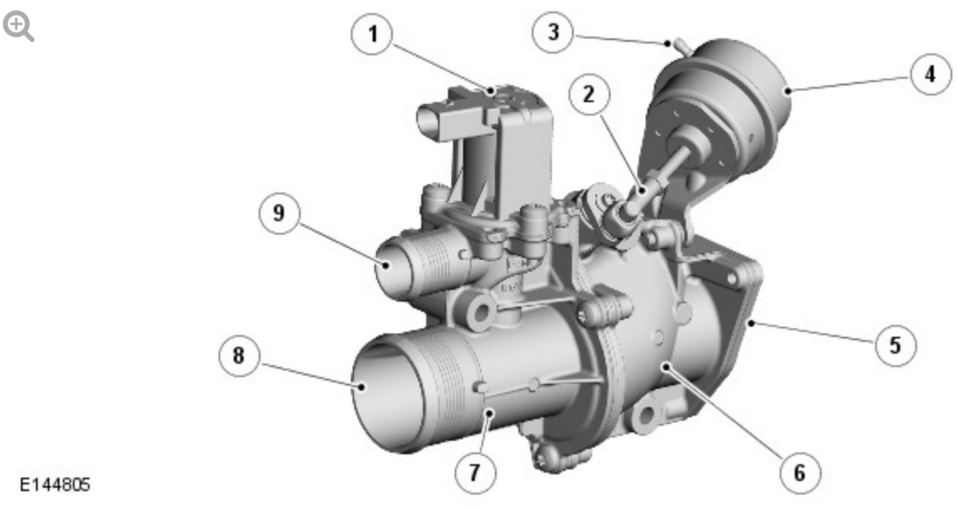
A second flange on the turbocharger provides for the attachment of the right down pipe of the exhaust system. Two screws secure the down pipe to the flange and a gasket seals the joint.

The compressor end of the turbocharger has two hose connections. The central connection provides a clean air supply from the air cleaner to the compressor. The second connection on the outside of the housing provides for the attachment of the charge air ducts.

A pipe on the center housing of the turbocharger provides for the connection of a bleed air pipe from the charge air duct connection with the electric throttle. The bleed air is used to pressurize the seals of the compressor stage to prevent oil leaking into the compressor housing when the turbocharger is not running.

Attached to the rear of the turbocharger is a turbine intake shut-off valve operated by a vacuum actuator. The ECM operates the turbine intake shut-off valve solenoid to apply and release the vacuum in the actuator, which opens the turbine intake shut-off valve when a vacuum is present. The turbine intake shut-off valve is closed when the system is operating in mono-turbocharger mode, diverting exhaust gases from the right exhaust manifold, via the exhaust cross-over duct to the left exhaust manifold. A position sensor in the turbine intake shut-off valve allows the ECM to monitor the shut-off valve for correct operation.

CHARGE AIR INTAKE VALVE (IF EQUIPPED)



ITEM	DESCRIPTION
1	Charge air recirculation solenoid
2	Charge air shut-off valve connecting rod
3	Vacuum connection
4	Vacuum actuator
5	Charge air outlet connection (to charge air coolers)
6	Charge air shut-off valve body
7	Charge air recirculation solenoid body
8	Charge air intake (from secondary turbocharger)
9	Charge air recirculation solenoid outlet connection (to primary turbocharger clean air duct)

The charge air intake valve is attached to a bracket on the front subframe and the cooling pack protector. The ECM uses the valve to control the bi-turbocharger operation of the turbochargers.

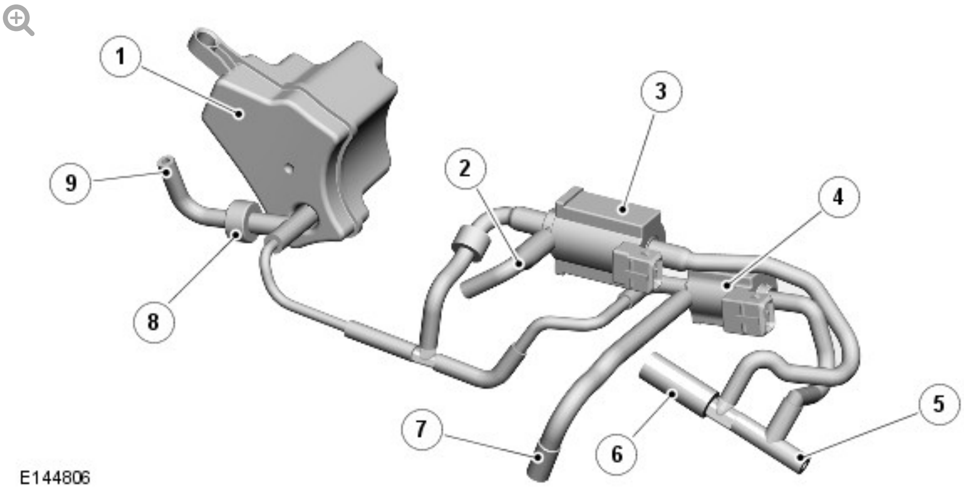
The charge air intake valve consists of a recirculation solenoid and a shut-off valve that are connected together by four screws and a gasket. The charge air recirculation solenoid controls the flow of air from the compressor of the secondary turbocharger to the clean air duct of the primary turbocharger. The charge air shut-off valve controls the flow of air from the compressor of the secondary turbocharger to the charge air coolers.

The charge air recirculation solenoid is operated by a power feed from the Engine Junction Box (EJB) and a ground connection with the ECM. When it is de-energized, the charge air recirculation solenoid is closed.

If the charge air recirculation solenoid develops a fault it will default to the closed position. In this position an overspeed of the primary turbocharger will occur. This overspeed is sensed by the Manifold Absolute Pressure (MAP) sensor which detects the increased pressure. The ECM registers the signals from the MAP sensor and operates the turbochargers in mono-turbo mode with restricted engine torque and Diagnostic Trouble Code (DTC)'s recorded.

The charge air shut-off valve is a flap valve operated by a vacuum actuator. The ECM operates the charge air solenoid to apply and release the vacuum in the actuator, which opens the valve when a vacuum is present. When the shut-off valve is open charge air from the secondary turbocharger flows through the valve to the charge air coolers in common with the air flow from the primary turbocharger.

VACUUM CONTROL SOLENOIDS (IF EQUIPPED)



ITEM	DESCRIPTION
1	Vacuum reservoir
2	Vacuum connection to turbine intake shut-off valve (secondary turbocharger)
3	Turbine intake shut-off valve solenoid
4	Charge air shut-off valve solenoid
5	Pipe from left air cleaner
6	Pipe to cooling bypass solenoid of Exhaust Gas Recirculation (EGR) system
7	Pipe to vacuum actuator of charge air shut-off valve
8	Non return valve
9	Pipe from vacuum pump

The turbine intake shut-off valve and charge air shut-off valve solenoids are installed on a bracket attached to the front of the Bank 2 cylinder head. The ECM operates the solenoids to apply and release the vacuum in the vacuum actuators of the turbine intake shut-off valve (located on the secondary turbocharger) and the charge air shut-off valve.

The vacuum for the actuators is produced by the vacuum pump on the rear of the Bank 1 cylinder head.
For additional information, refer to: Engine (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation) /

Engine (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).

The solenoids are normally-closed valves installed in the vacuum line between the vacuum reservoir and their related vacuum actuator. Each solenoid is also connected to a line from the left air cleaner, which allows air into the vacuum lines to the actuators when the solenoids are de-energized.

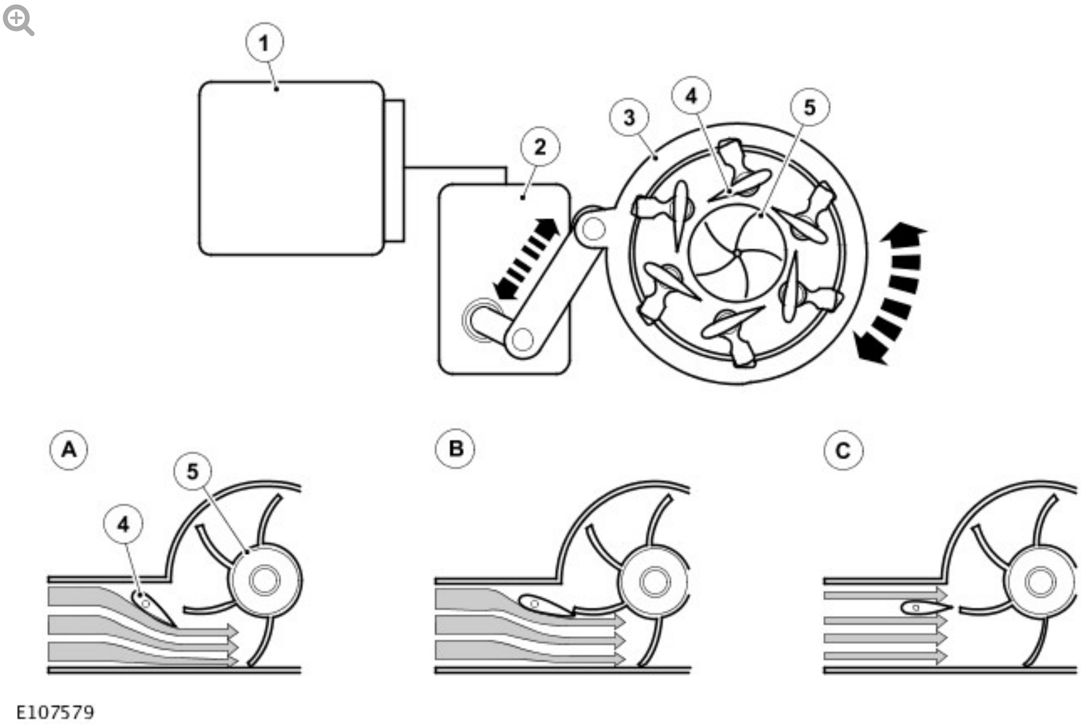
OPERATION

PRINCIPLES OF VARIABLE VANE OPERATION

The turbine wheel of each turbocharger uses the engine exhaust gases to drive the compressor wheel. The compressor wheel draws in clean air, which is supplied to the engine cylinders in a compressed form.

The primary, VGT turbocharger allows the optimum inlet geometry (inlet area and flow angle) to be used over a wide range of engine operating conditions. This allows a rapid speed of response and higher charge air pressures at low engine speeds. The variable vane angle determines both the inlet area as well as the flow angle, as controlled by the ECM via the VGT actuator. The variable vanes allow efficient use of the exhaust gas energy which in turn improves turbocharger and engine efficiency.

Schematic Diagram - Variable Vane Operation



A = LOW ENGINE SPEED; B = INTERMEDIATE ENGINE SPEED; C = MAXIMUM ENGINE SPEED.

ITEM	DESCRIPTION
1	Engine Control Module (ECM)
2	Variable Geometry Turbine (VGT) vane actuator
3	Turbine housing
4	Variable vanes
5	Compressor wheel

Low Engine Speed: At low engine speeds the volume of exhaust gas is low so the vanes are moved towards the closed position to reduce the turbine inlet area. This reduction causes an increase in the gas velocity into the turbine wheel thereby increasing wheel speed and charge air pressure.

Intermediate Engine Speed: As the engine speed increases and the volume of exhaust gas increases the vanes are moved towards the open position to increase the turbine inlet area and maintain the gas velocity.

Maximum Engine Speed: At maximum engine speed the vanes are almost fully open maintaining the gas velocity into the turbine wheel.

When the vehicle is driven at high altitudes the ambient pressure reduces causing the compressor wheel to rotate faster to achieve the same charge air pressure. To prevent the turbine wheel from over-speeding under these conditions, the ECM protects the turbocharger by opening the vanes further to reduce the turbine wheel speed. This is known as the altitude margin of the turbocharger, which occurs at altitudes above 2000 meters. Below 2000 meters altitude compensation is not necessary. The ECM uses an internal barometric pressure sensor to monitor altitude.

The maximum position of the turbocharger vanes (fully open) is also an emergency default position in the event of an electrical fault. This lowers the chance of engine damage due to excessive charge air pressure.

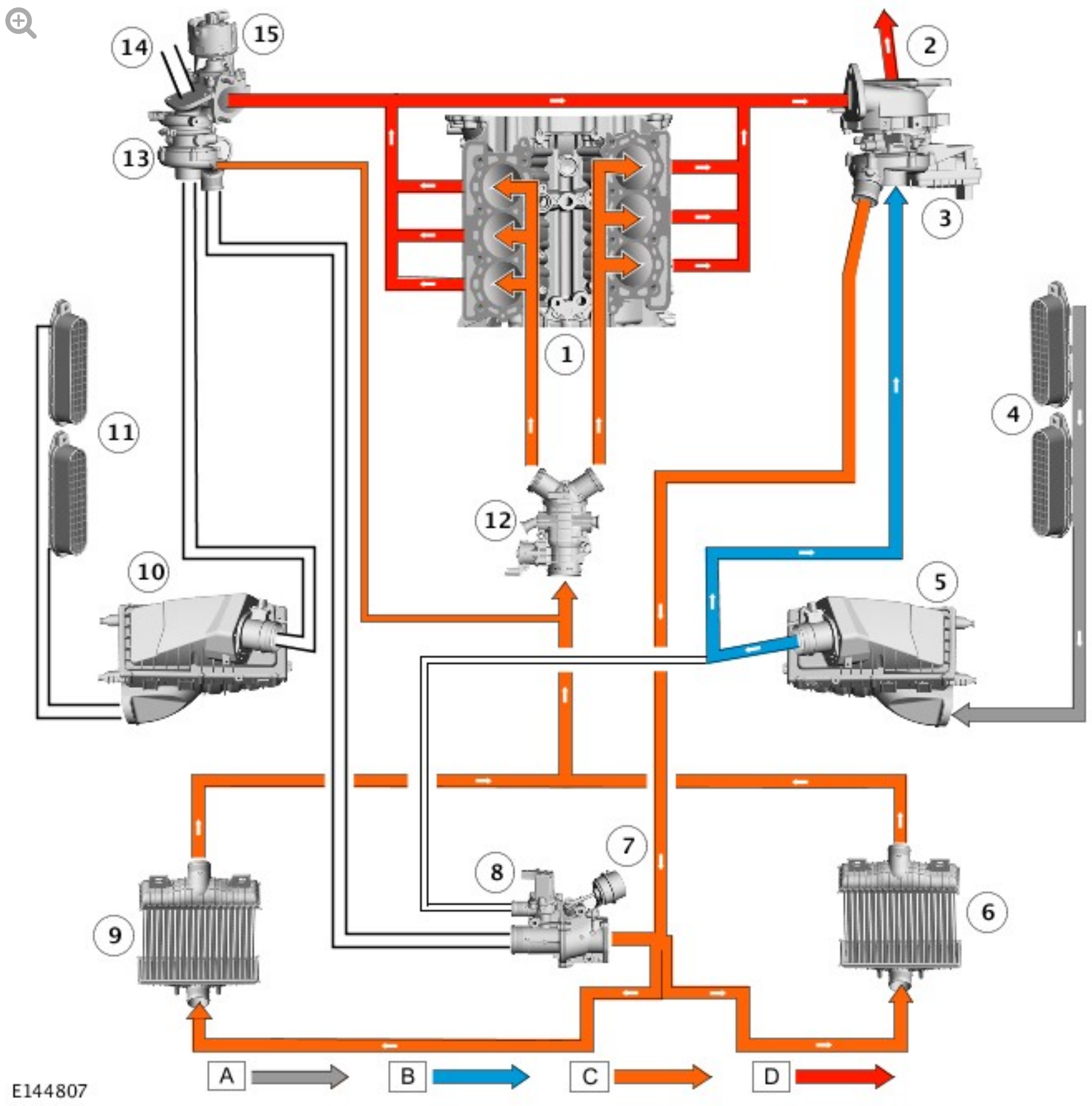
DUAL MODE BOOSTING (IF EQUIPPED)

CAUTION:

> Ensure both ends of the pipe are securely connected to the secondary turbo and the air intake system to prevent damage to the turbo components.

The dual mode boosting system comprises two turbochargers and software within the ECM. The two turbochargers can operate in two modes; mono turbocharger operation or bi-turbocharger operation.

MONO TURBOCHARGER OPERATION



A = DIRTY AIR; B = CLEAN AIR; C = CHARGE AIR; D = EXHAUST GASES.

ITEM	DESCRIPTION
1	Engine
2	Exhaust system left intake pipe
3	Primary turbocharger
4	Left intake grills
5	Left air cleaner
6	Left charge air cooler
7	Charge air shut-off valve
8	Charge air recirculation solenoid

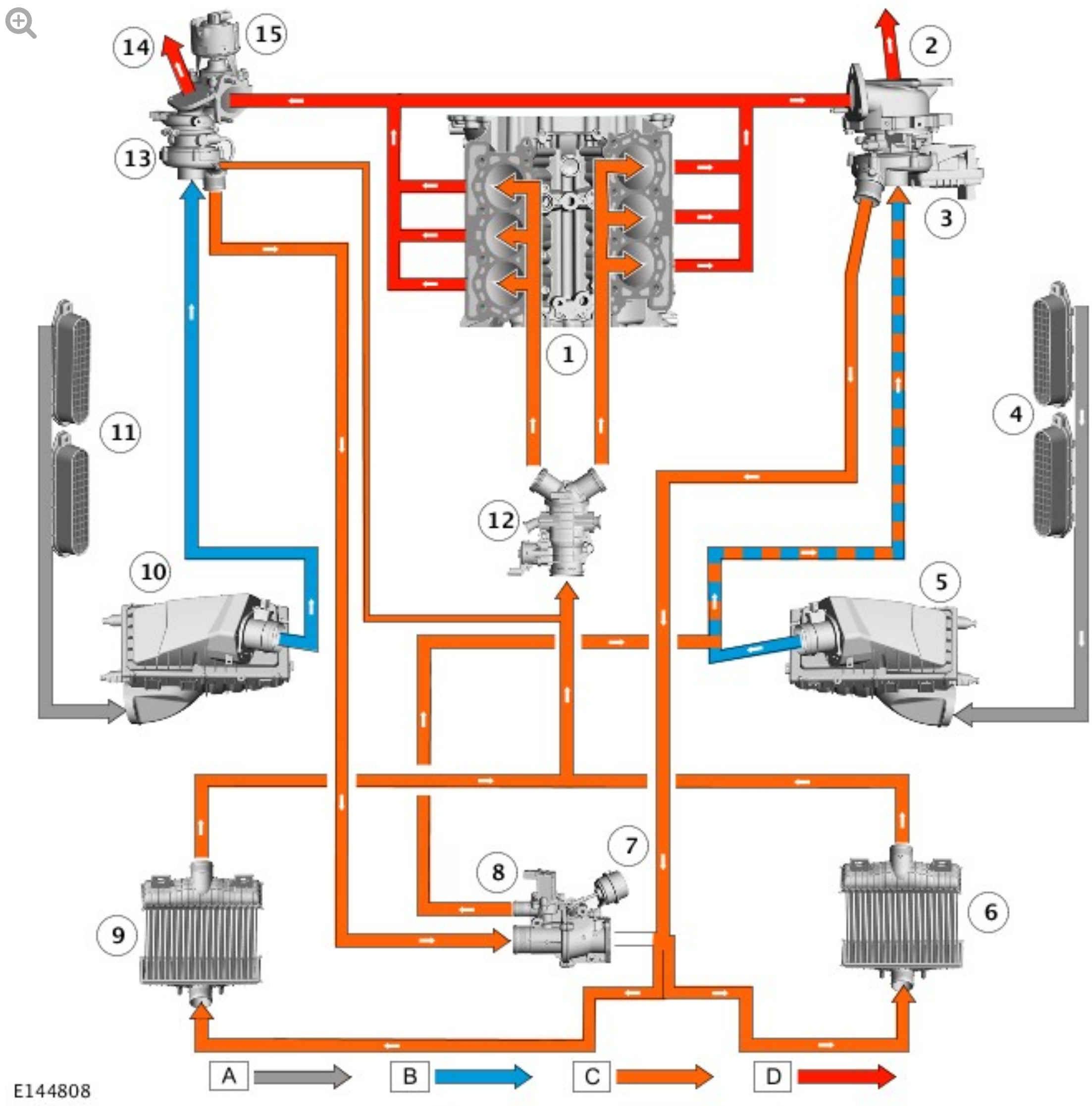
9	Right charge air cooler
10	Right air cleaner
11	Right intake grills
12	Electric throttle
13	Secondary turbocharger
14	Exhaust system right intake pipe
15	Turbine shut-off valve

Fresh air is drawn through the left air cleaner and the Mass Air Flow (MAF) sensor to the primary turbocharger compressor. The compressed air then passes through the charge air coolers and electric throttle to the intake manifolds.

The turbine shut-off valve on the secondary turbocharger is closed and therefore exhaust gases are unable to operate the secondary turbocharger turbine. In this condition all charge air pressure is produced by the primary turbocharger using the exhaust gases from all six cylinders.

If the engine idles for more than 3 minutes, the secondary turbocharger is actuated to keep the turbocharger shaft rotating, to prevent oil from leaking into the turbine housing. This is achieved by periodically opening the turbine intake shut-off valve to operate the turbocharger.

BI-TURBOCHARGER SWITCHING



A = DIRTY AIR; B = CLEAN AIR; C = CHARGE AIR; D = EXHAUST GASES.

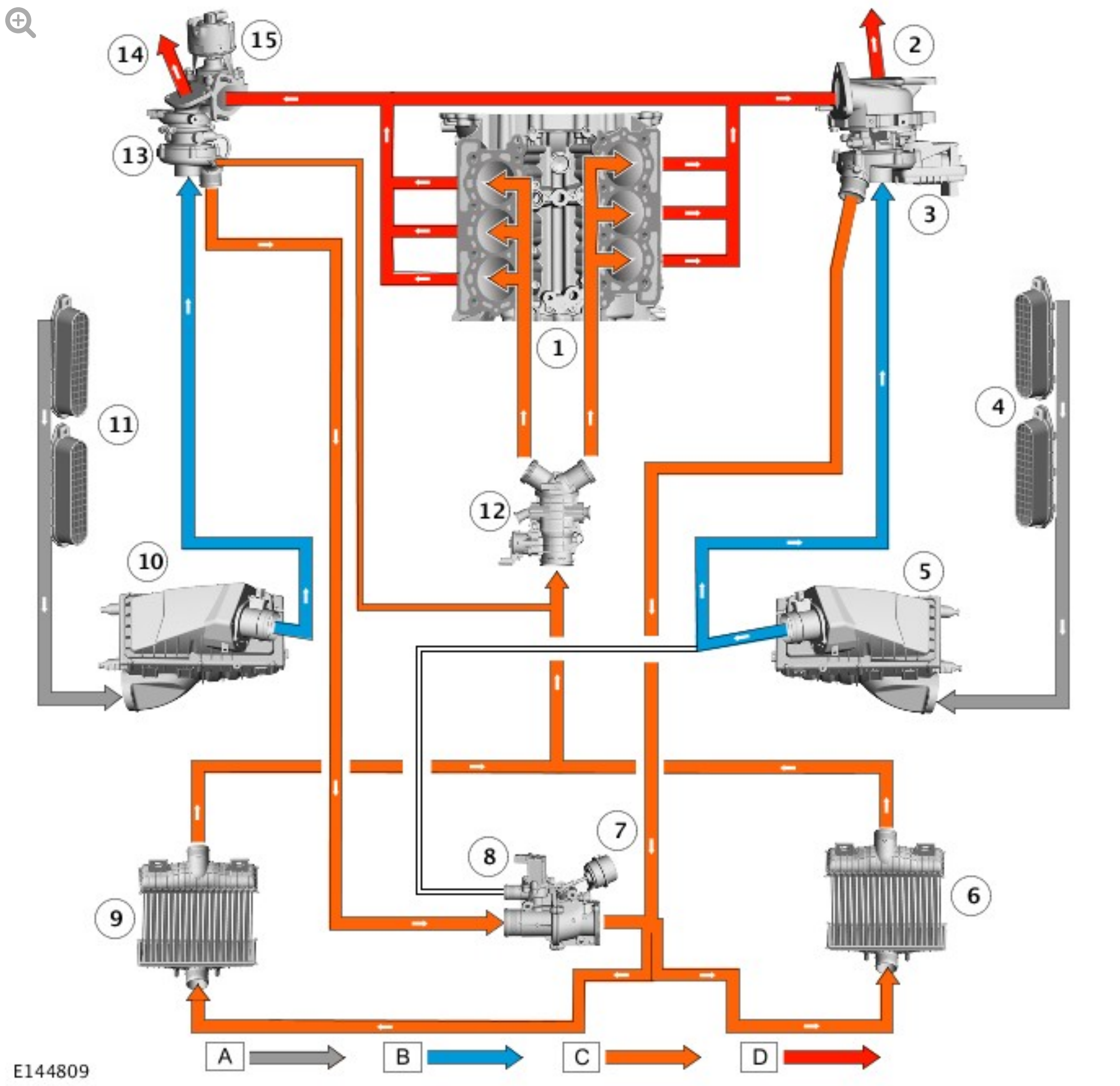
ITEM	DESCRIPTION
1	Engine
2	Exhaust system left intake pipe
3	Primary turbocharger
4	Left intake grills
5	Left air cleaner
6	Left charge air cooler
7	Charge air shut-off valve
8	Charge air recirculation solenoid

9	Right charge air cooler
10	Right air cleaner
11	Right intake grills
12	Electric throttle
13	Secondary turbocharger
14	Exhaust system right intake pipe
15	Turbine shut-off valve

When the engine operating parameters approach the limits (approximately 2800 RPM under load) of the primary turbocharger, the ECM begins the switch to parallel bi-turbocharger operation. The secondary turbocharger is brought into operation by the opening of the turbine shut-off valve which allows exhaust gases to flow through the turbine. Fresh air is then drawn through the right air cleaner and the Mass Air Flow and Temperature (MAFT) sensor to the secondary turbocharger compressor.

Initially, the secondary turbocharger does not produce a charge air pressure to equal that of the primary turbocharger. Therefore, the charge air from the secondary turbocharger is fed via the recirculation solenoid into the clean air duct of the primary turbocharger.

BI-TURBOCHARGER OPERATION



A = DIRTY AIR; B = CLEAN AIR; C = CHARGE AIR; D = EXHAUST GASES.

ITEM	DESCRIPTION
1	Engine
2	Exhaust system left intake pipe
3	Primary turbocharger
4	Left intake grills
5	Left air cleaner
6	Left charge air cooler
7	Charge air shut-off valve
8	Charge air recirculation solenoid

9	Right charge air cooler
10	Right air cleaner
11	Right intake grills
12	Electric throttle
13	Secondary turbocharger
14	Exhaust system right intake pipe
15	Turbine shut-off valve

As the secondary turbocharger charge air pressure increases, the recirculation solenoid is closed and the charge air shut-off valve opened. This increases the charge air pressure from the secondary turbocharger and combines it with the charge air from the primary turbocharger.

The ECM maintains the turbochargers in bi-turbocharger mode until it determines that the engine operating parameters no longer require it. The ECM then switches the turbochargers back to mono turbocharger operation.

SHUT-OFF VALVE CONTROL

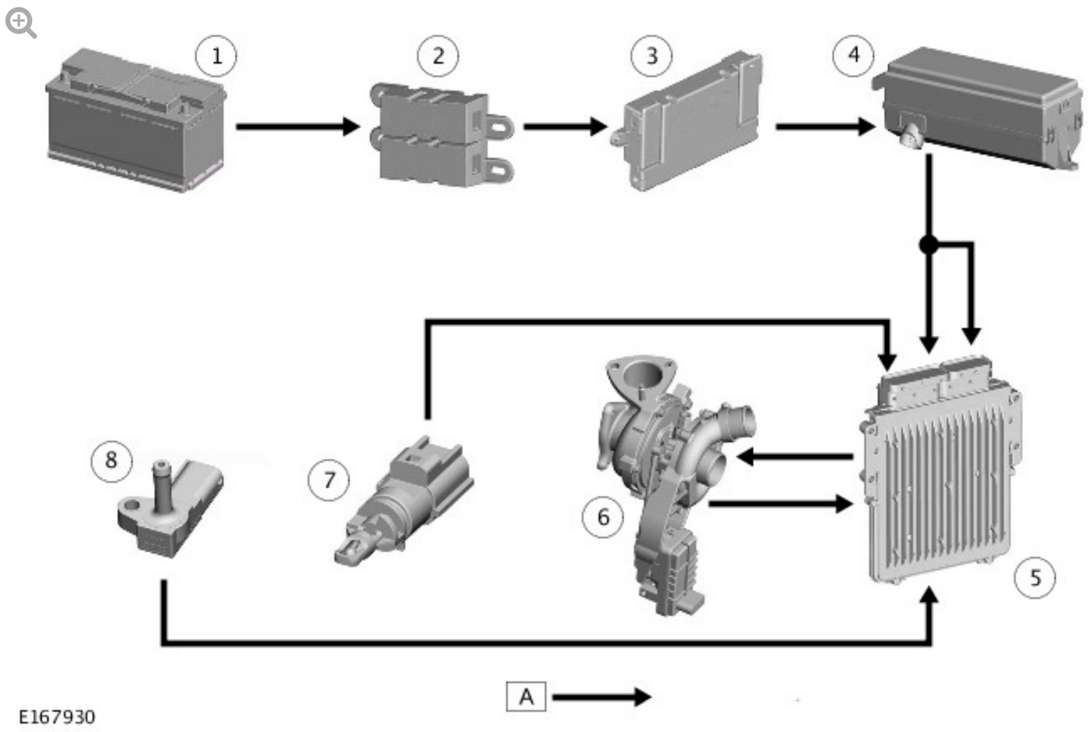
Turbine intake shut-off valve: If vacuum is released from the turbine intake shut-off valve it will default to the closed position. The turbine intake shut-off valve position sensor will inform the ECM, which will initiate mono-turbo mode, restricts engine performance and record a DTC.

Charge air shut-off valve: If vacuum is released from the charge air shut-off valve it will default to the closed position. In the default position, the ECM initiates mono-turbo operation and restricts engine torque. A DTC will also be recorded in the ECM.

CONTROL DIAGRAM - SHEET 1 OF 2 - VEHICLES WITH SINGLE TURBOCHARGER

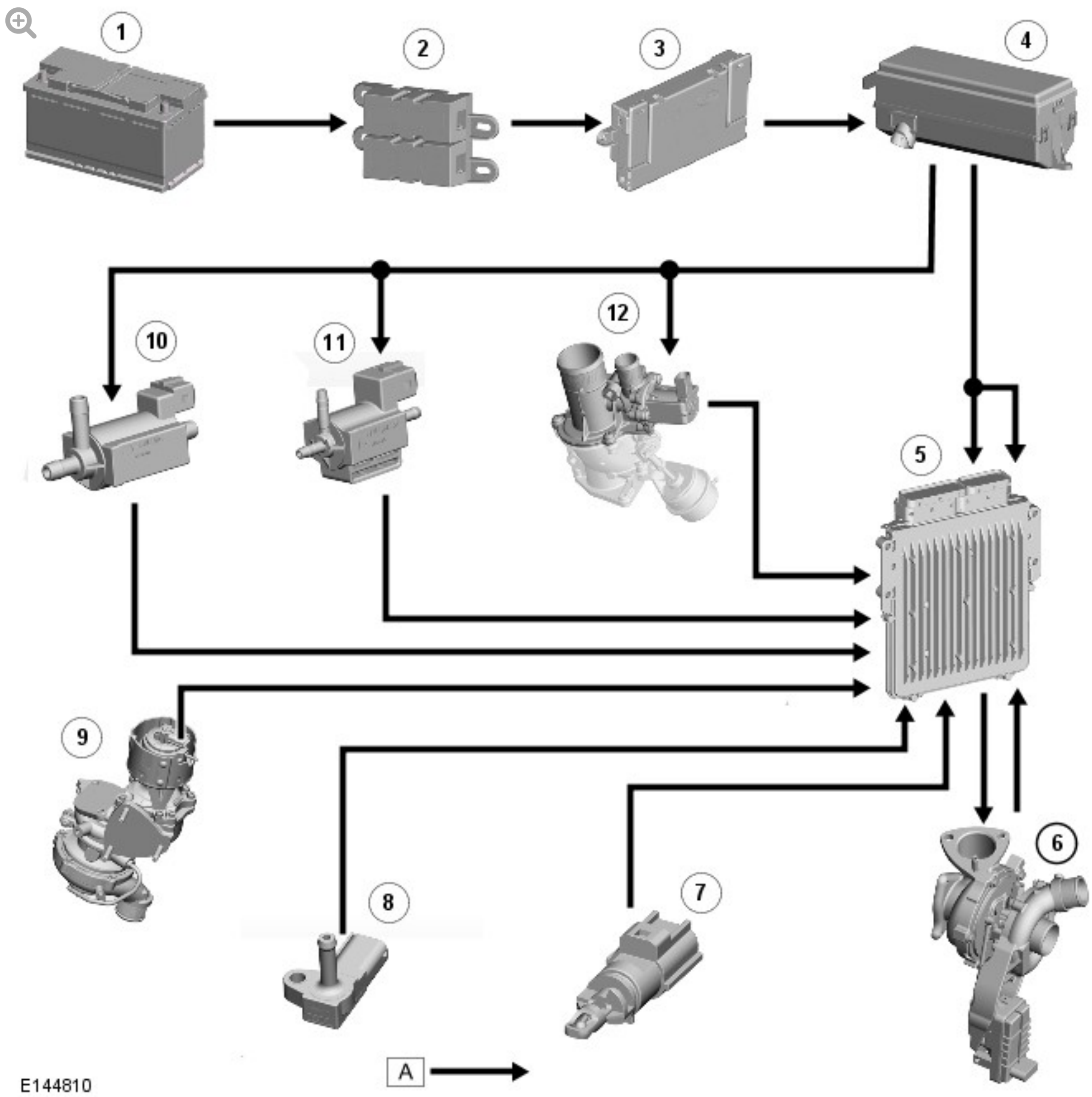
NOTE:

A = Hardwired.



ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Engine Junction Box (EJB)
5	Engine Control Module (ECM)
6	Variable Geometry Turbine (VGT) vane actuator
7	Charge air temperature sensor
8	Charge air pressure sensor

CONTROL DIAGRAM - SHEET 2 OF 2 - VEHICLES WITH TWIN TURBOCHARGER



ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Engine Junction Box (EJB)
5	Engine Control Module (ECM)
6	Variable Geometry Turbine (VGT) vane actuator
7	Charge air temperature sensor
8	Charge air pressure sensor
9	Turbine intake shut-off valve position sensor
10	Turbine intake shut-off valve solenoid

11	Charge air solenoid
12	Charge air recirculation solenoid